Sharpness of the $k/4$ Triangle Constant for Contractions

FA/Banach run packet

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Source Question

Bourin and Lee [1, Corollary 4.4] prove that, for contractions $A_1, \dots, A_k \in M_n$, $k > 1$,

$$\left| \sum_{j=1}^{k} A_j \right| \le \frac{k}{4} I + \sum_{j=1}^{k} |A_j|.$$

They prove that the constant $k/4$ is sharp for even k, and conjecture that it is also sharp for odd $k > 1$. The abstract singles out the case $k = 3$, where the conjectured optimal constant is $3/4$.

Result

Theorem. *For every integer $k > 1$, the constant $k/4$ in the Bourin–Lee triangle inequality is optimal. More precisely, over complex matrices, there are rank-one contractions $A_0, \dots, A_{k-1} \in M_2(\mathbb{C})$ such that any inequality*

$$\left| \sum_{j=0}^{k-1} A_j \right| \le cI + \sum_{j=0}^{k-1} |A_j|$$

forces $c \ge k/4$.

Proof. Let e_1, e_2 be the standard basis of $\mathbb{C}^2$, let $\zeta = \exp(2\pi i/k)$, and put

$$v_j = \frac{1}{2} e_1 + \frac{\sqrt{3}}{2} \zeta^j e_2, \qquad u = e_1, \qquad A_j = u v_j^*$$

for $j = 0, \dots, k-1$. Each v_j is a unit vector, so each A_j is a rank-one contraction. Moreover

$$A_j^* A_j = v_j v_j^*, \qquad |A_j| = v_j v_j^*.$$

Since $k > 1$, $\sum_{j=0}^{k-1} \zeta^j = 0$. Hence

$$\sum_{j=0}^{k-1} A_j = u \left(\sum_{j=0}^{k-1} v_j \right)^* = \frac{k}{2} e_1 e_1^*,$$

and therefore

$$\left| \sum_{j=0}^{k-1} A_j \right| = \frac{k}{2} e_1 e_1^*.$$

On the other hand, the off-diagonal terms in $\sum_j v_j v_j^*$ vanish by the same roots-of-unity identity, giving

$$\sum_{j=0}^{k-1} |A_j| = \sum_{j=0}^{k-1} v_j v_j^* = \begin{pmatrix} k/4 & 0 \\ 0 & 3k/4 \end{pmatrix}.$$

If the displayed inequality with constant c holds for this family, then taking the quadratic form at e_1 gives

$$\frac{k}{2} \leq c + \frac{k}{4}.$$

Thus $c \geq k/4$. Since the Bourin–Lee corollary proves the inequality with $c = k/4$, the constant is optimal. $\square$

Corollary. *The odd-k conjecture in [1] is true. In particular, for three contractions the constant* $3/4$ *cannot be decreased.*

Real Matrices

For odd $k \geq 3$, the same argument has a real 3×3 version. Let

$$v_j = \frac{1}{2} e_1 + \frac{\sqrt{3}}{2} \left(\cos \frac{2\pi j}{k}\, e_2 + \sin \frac{2\pi j}{k}\, e_3 \right), \qquad A_j = e_1 v_j^T.$$

The vertices of the regular k-gon have zero sum, and

$$\sum_{j=0}^{k-1} |A_j| = \operatorname{diag}(k/4, 3k/8, 3k/8), \qquad \left| \sum_{j=0}^{k-1} A_j \right| = \frac{k}{2} e_1 e_1^T.$$

Evaluating at e_1 again forces $c \geq k/4$.

Scope

This packet settles the optimality of the additive constant in the triangle inequality for contractions. It does not address separate sharpness questions for other eigenvalue inequalities in [1].

References

[1] J.-C. Bourin and E.-Y. Lee, *Diagonal and off-diagonal blocks of positive definite partitioned matrices*, arXiv:2307.02034; Linear Algebra Appl. 684 (2024), 87–100.